

Density-Based Traffic Signal Control

User Guide — Navigating and Using the Application

Operator manual

2026-06-15

Contents

1	Introduction	1
1.1	Launching	1
2	The Main Window at a Glance	1
2.1	Docking and the View menu	1
3	The Cameras Panel	1
3.1	What you see on each feed	2
4	The Left Toolbar — Core Actions	2
4.1	Add source	2
4.1.1	Live camera (RTSP)	2
4.1.2	Proxy video file	3
4.2	Remove source	3
4.3	Display settings	3
4.4	Edit region (ROI)	3
4.4.1	Using the ROI editor	3
4.5	Experimental settings	3
5	Monitoring Panels	4
5.1	System Metrics	4
5.2	Charts	4
5.3	Logs	4
6	The Settings Menu — Configurations	4
7	Signal-Control Behaviour (What the App Decides)	4
7.1	Tuning the emergency thresholds	5
8	Typical Workflows	5
8.1	Run a video-file demo (no hardware)	5
8.2	Set up the live intersection	5
8.3	Improve performance on a modest GPU	5
9	Tips and Troubleshooting	6

1 Introduction

This guide explains how to operate the desktop control panel after it has been installed (see the Installation Guide). The application detects vehicles in up to four camera feeds, decides which of the two traffic lights (the *vertical* lane and the *horizontal* lane) should be green, and publishes those decisions over MQTT. It also lets emergency vehicles (ambulances, firetrucks) preempt the signal.

1.1 Launching

From the repository root, with the virtual environment active:

```
source env/bin/activate
python gui/main.py
```

The main window opens with the title **Control Panel**. All settings are remembered between runs, so on later launches the app reopens with your last cameras, layout, and options.

2 The Main Window at a Glance

The window is built from a menu bar, a vertical toolbar on the left, several dockable panels, and a status bar along the bottom.

Area	Purpose
Menu bar (top)	Settings and View menus: load/save configs, reset, toggle panels.
Left toolbar	Add/remove cameras, display settings, edit ROI, experimental settings.
Cameras panel	2×2 grid of live video with detections and overlays.
System Metrics panel	Clock, CPU/RAM/GPU usage, uptime, FPS, latency breakdown.
Logs panel	Running list of status messages.
Delay Metrics Chart	Capture / inference / display latency over time.
System Metrics Chart	CPU / RAM / GPU usage over time.
Status bar	Short transient messages about the last action.

2.1 Docking and the View menu

Every panel is a dock that can be moved, floated, resized, or closed. Drag a panel’s title bar to rearrange it, or drag it out of the window to float it. The **View** menu lists each panel with a checkbox to show or hide it, plus **Reset layout** to restore the standard arrangement. Your layout is saved automatically and restored on the next launch.

3 The Cameras Panel

The Cameras panel shows a 2×2 grid of feeds. The four slots are filled in the order your cameras were added: slots 1–2 on the top row, slots 3–4 on the bottom row. An empty slot reads “*Missing camera source.*” Drag the splitter bars between feeds to resize them.

3.1 What you see on each feed

Depending on the display settings (§4.3), each feed can show:

- **Bounding boxes** around detected objects, colour-coded: green = vehicle, yellow = ambulance, red = firetruck.
- **Labels** (class name and confidence) when annotations are on.
- **An info overlay** (top-left) with camera name, location, and an estimated congestion ratio.
- **A traffic-signal indicator** (top-right) showing this lane's current red / yellow / green state.
- **The region-of-interest outline** when ROI display is enabled.

4 The Left Toolbar — Core Actions

The toolbar holds the five everyday actions. Each is described below.

4.1 Add source

Opens a dialog to register a new camera. *Which* dialog appears depends on the **Use videos** option (§4.5):

4.1.1 Live camera (RTSP)

When **Use videos** is off, the **Add Camera Source** form requests:

- **Camera Name** and **Location** — free text for identification.
- **Max cap** — the lane's vehicle capacity, used for the congestion estimate.
- **IP Address** — validated as a dotted IPv4 address (e.g. 192.168.1.64).
- **Username / Password** — camera login (password is masked).
- **RTSP Port** — defaults to 554.
- **Channel** — defaults to 101 (main stream of channel 1 on Hikvision-style cameras).
- **Direction** — **Vertical** or **Horizontal**; this assigns the camera to one of the two traffic lights.

The stream URL is assembled as `rtsp://user:pass@ip:port/Streaming/Channels/channel`. All fields are required; click **Add** to save. The detector restarts automatically with the new camera.

4.1.2 Proxy video file

When **Use videos** is on, the **Add Proxy Video Source** form requests a **Camera Name**, **Location**, **Max cap**, **Direction**, and a **Video File** chosen with **Browse** (.mp4, .avi, .mkv, .mov). This is the recommended way to demo or test the system without live cameras. Video sources loop continuously.

4.2 Remove source

Opens **Remove Camera Source(s)**, showing two checklists side by side: **Camera Sources** (live) and **Proxy Camera Sources** (video). Tick any combination across both lists and click **Remove Selected**. The detector restarts with the remaining sources. **Cancel** discards the selection.

4.3 Display settings

Opens **Change Display Settings**, which controls only what is *drawn*; changes apply immediately without restarting the detector. Three groups:

- **Bounding Box Display** — show/hide boxes for **Firetrucks**, **Ambulances**, and **Vehicles** independently. Enable firetruck and ambulance boxes when you want to visually confirm emergency detections.
- **On-Screen Display (OSD)** — toggle each overlay element: **Name**, **Location**, **Estimated Congestion**, **Annotations** (class/confidence labels), **Region of Interests** (the ROI outline), and **Traffic Phase** (the red/yellow/green indicator).
- **Logs** — the log verbosity level recorded in the Logs panel: **Debug**, **Info**, or **Errors**.

Click **Save** to apply, **Cancel** to discard.

4.4 Edit region (ROI)

A region of interest restricts detection to the part of the frame that is actual road, which improves both accuracy and speed. Click **Edit Region**, pick a camera from the dropdown, and click **OK** to open the ROI editor.

4.4.1 Using the ROI editor

The editor shows a snapshot from that camera with a blue quadrilateral overlay:

- **Drag a corner handle** (the light-blue squares) to reshape the region.
- **Drag inside the polygon** to move the whole region.
- The region is clamped to the image bounds.
- Click **Save** to store the ROI (kept as normalised coordinates so it survives resolution changes), or **Cancel** to discard.

Draw the quadrilateral tightly around the lane(s) feeding the intersection. After saving, enable **Region of Interests** in the display settings to see the outline on the live feed. Detection thereafter runs only inside this region.

4.5 Experimental settings

Opens **Change Experimental Settings**. Saving here *restarts* the detector. Options:

- **Use videos** — switch the whole app between live RTSP cameras and local video files. This determines which source list is active and which **Add source** dialog appears.
- **Tile Width / Tile Height** — the resolution each feed is processed at. Smaller tiles run faster (useful on a modest GPU such as an RTX 2060); larger tiles detect smaller/farther vehicles more reliably.
- **Model confidence** — the detection threshold, adjustable from 0.00 to 1.00 via the slider or the spin box (they stay in sync). Lower values detect more but risk false positives; the default is 0.30.

5 Monitoring Panels

5.1 System Metrics

A compact readout updated once per second: current date and time, **CPU**, **RAM** and **GPU** utilisation bars, **Uptime**, the processing **FPS**, and a latency breakdown — **Total Delay** and its **Cam**, **Inf** and **Disp** components (capture, inference, and display time in milliseconds). FPS is derived as 1/total delay.

5.2 Charts

Two live charts plot the same data over time so you can spot trends:

- **Delay Metrics Chart** — camera, inference, and display latency.
- **System Metrics Chart** — CPU, RAM, and GPU usage.

Both keep a rolling window of recent samples.

5.3 Logs

A scrolling, read-only history of status messages (most recent first), mirroring what briefly appears in the status bar — camera additions/removals, detector restarts, settings saves, and errors.

6 The Settings Menu — Configurations

Configurations bundle your cameras, display options, experimental options, and window layout into a single JSON file, so you can save and reload complete setups (e.g. one for the live intersection, one for a video demo).

- **Load settings** — import a configuration from a `.json` file and restart the detector with it. Ready-made files live in `configs/` (`all_cameras.json` for the live cameras, `proxy_cameras.json` / `focus_first_camera.json` for video demos).
- **Save settings** — persist the current state to the app’s internal store (also happens automatically on most actions).
- **Save settings as** — export the current state to a `.json` file you choose.
- **Reset to default settings** — restore default display, experimental and layout settings (your camera list is kept).
- **Reset camera data** — clear all configured cameras.
- **Quit** — close the application.

7 Signal-Control Behaviour (What the App Decides)

While running, the app continuously counts vehicles per direction and drives the two lights. Understanding this helps interpret the on-screen signal indicator.

- **Density rule** — the direction with more vehicles is favoured for green; the other is held red.
- **Minimum green** — a newly granted green is held briefly before it can switch, preventing rapid flicker when counts are close.
- **Maximum green** — a direction cannot hold green indefinitely; after the cap, the other direction is given a turn.
- **Yellow transition** — the losing direction shows yellow before going red, then its partner turns green.
- **Emergency preemption** — a confirmed ambulance or firetruck forces its lane green immediately, bypassing the minimum-green hold and staying green while the emergency is present. A detection is only “confirmed” once it clears the emergency confidence threshold for several consecutive cycles, so a single noisy frame cannot trigger it.

7.1 Tuning the emergency thresholds

The two emergency knobs are *not* in the experimental-settings dialog; they are set in the configuration file’s `experimental_settings` block and applied when the config is loaded:

```
"emergency_confidence": 0.55,    // min confidence for an emergency to count
"emergency_frames": 5           // consecutive cycles required to confirm
```

Raise `emergency_confidence` or `emergency_frames` if you see false preemptions; lower them if genuine emergency vehicles are reacting too slowly. To watch this live, enable the **Ambulances** and **Firetrucks** bounding boxes in the display settings.

8 Typical Workflows

8.1 Run a video-file demo (no hardware)

1. **Settings** → **Load settings** and pick `configs/proxy_cameras.json`, *or...*
2. open **Experimental settings**, tick **Use videos**, **Save**; then use **Add source** to add one or more video files.
3. Set a **Direction** for each so both vertical and horizontal lanes are represented.
4. Optionally draw an ROI per source (§4.4).
5. Watch the feeds, the signal indicators, and the metrics panels.

8.2 Set up the live intersection

1. Ensure **Use videos** is off in **Experimental settings**.
2. **Add source** for each camera with its IP, credentials, and **Direction** (or load `configs/all_cameras.json`).
3. Draw an ROI for each camera over the relevant lanes.
4. Tune **Model confidence** and tile size for your GPU.
5. Confirm the signal indicators react and that the ESP32 lights follow (see the Installation Guide's smoke test).
6. **Settings** → **Save settings as** to store this deployment.

8.3 Improve performance on a modest GPU

1. Lower **Tile Width/Height** in **Experimental settings**.
2. Draw tight ROIs so inference covers less area.
3. Watch the **Inf Delay** in System Metrics and the Delay Metrics Chart; aim for a stable, low inference time and acceptable FPS.

9 Tips and Troubleshooting

- **A feed says “Missing camera source”** — that grid slot has no camera; add one, or the stream/file could not be opened (check the IP, credentials, network, or file path).
- **Detector seems stuck after changing options** — adding/removing sources and saving experimental settings restart the detector; give it a moment to reload the model.
- **Boxes/overlays not showing** — check the corresponding toggles in **Display settings**; these apply instantly.
- **Too many/few detections** — adjust **Model confidence**.
- **Detections outside the road** — draw or tighten the ROI.
- **Lost your layout** — **View** → **Reset layout**.
- **Want a clean slate** — **Settings** → **Reset to default settings** (keeps cameras) or **Reset camera data** (clears cameras).